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Aggregate Demand and Aggregate Supply

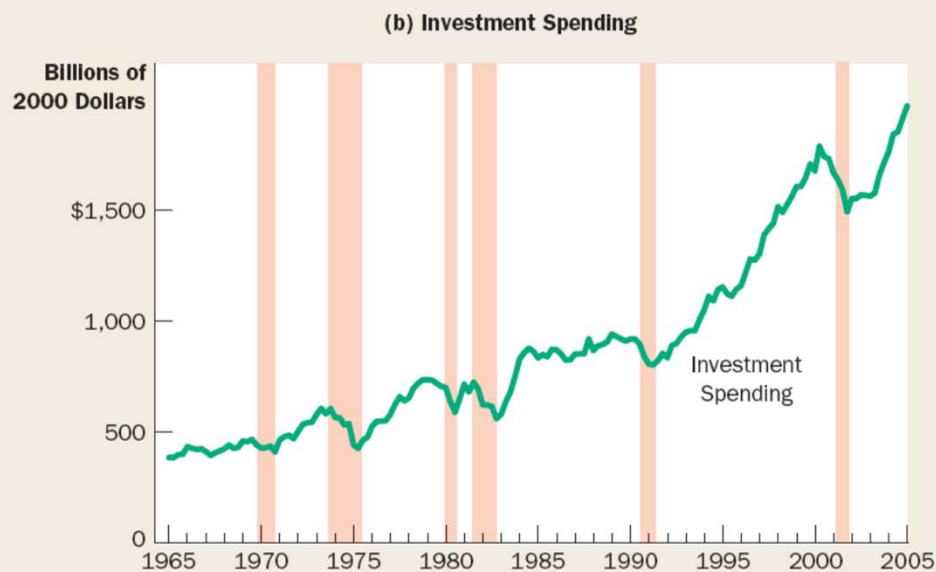
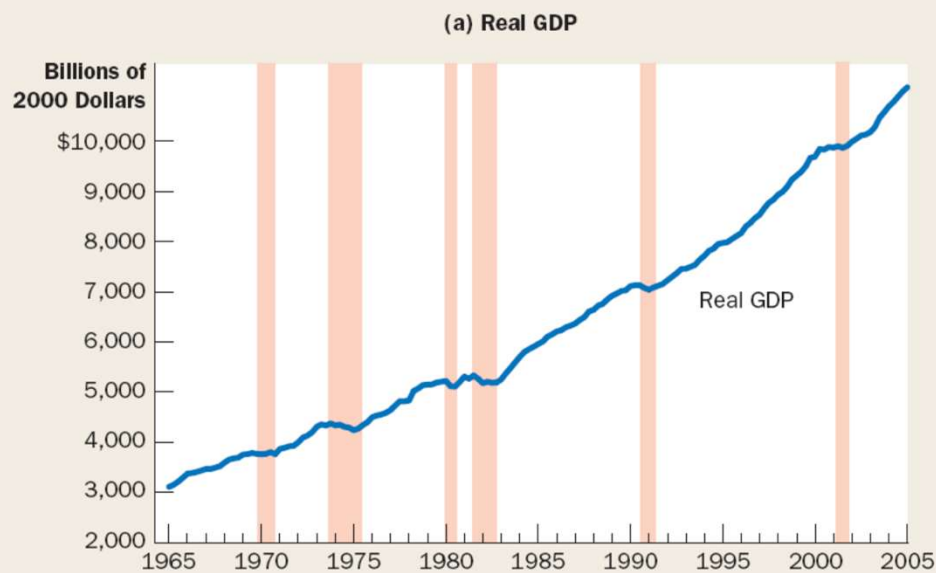


FIGURE 1

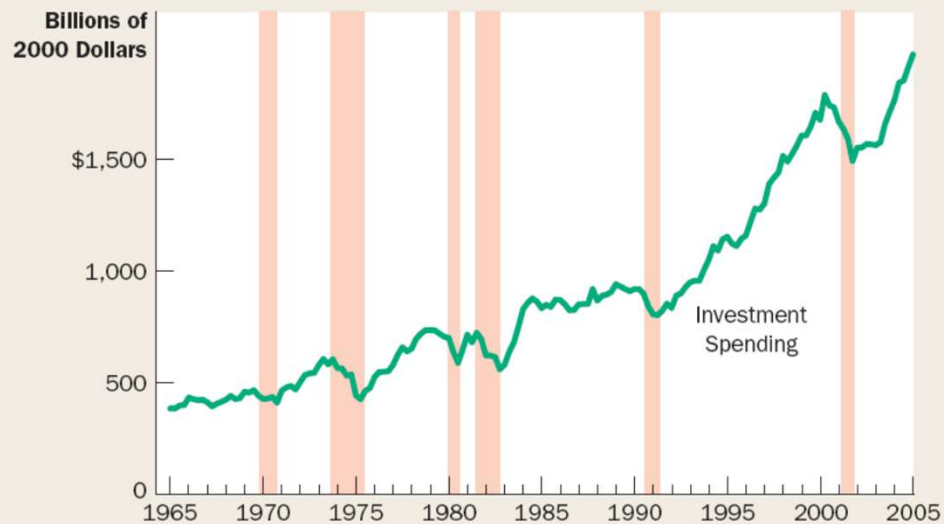
A Look at Short-Run Economic Fluctuations

This figure shows real GDP in panel (a), investment spending in panel (b), and unemployment in panel (c) for the U.S. economy using quarterly data since 1965. Recessions are shown as the shaded areas. Notice that real GDP and investment spending decline during recessions, while unemployment rises.

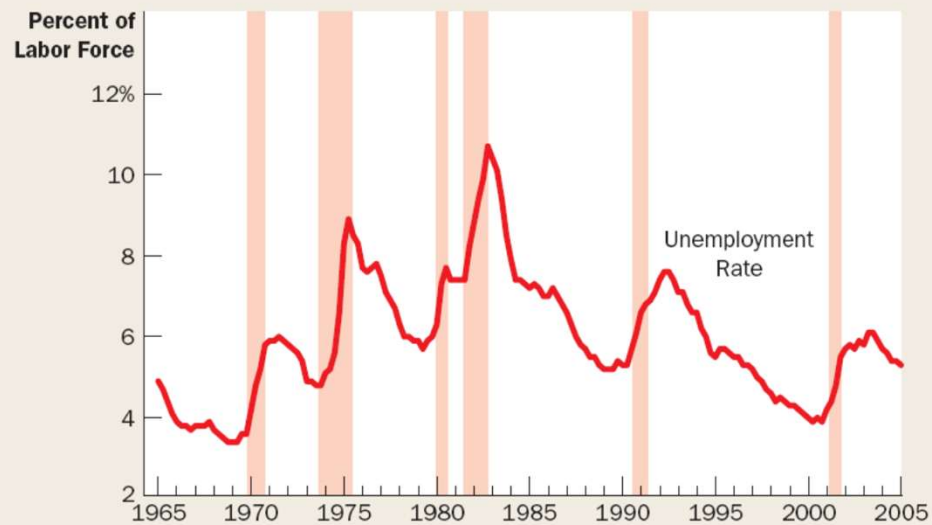
Source: U.S. Department of Commerce; U.S. Department of Labor.



(b) Investment Spending



(c) Unemployment Rate





Economic activity fluctuates from year to year. In most years, the production of goods and services rises. Because of increase in labour force, increase in the capital stock, and advances in technological knowledge, the economy can produce more and more over time. This growth allows everyone to enjoy a higher standard of living.



In some years this normal growth does not occur. Firms find themselves unable to sell all of the goods and services they have to offer, so they cut back the production. Unemployment rises and factories are left idle. With the economy producing fewer goods and services, real GDP and other measures of income fall. Such a period of falling incomes and rising unemployment is called a **recession** if it is relatively mild and **depression** if it is more severe.



Falling output and rising prices is known as stagflation.



short-run economic fluctuations: The Assumption of Classical Economies

Classical Dichotomy: is the separation of variables into real variables (those that measures quantities or relative prices) and nominal variables (those measured in terms of money).

Classical Dichotomy arises because, in classical theory, changes in the money supply do not influence real variables. This irrelevance of money for real variables is called **monetary neutrality**.



The Reality of Short-Run Fluctuations

Do these assumptions of classical macroeconomic theory apply to the world in which we live? No

Most economists believe that classical theory describes the world in the long run but not in the short run.

In the short run, real and nominal variables are highly intertwined, and changes in the money supply can temporarily push real GDP away from its long-run trend.

Even the classical economists themselves, such as David Hume, realized that classical economic theory did not hold in the short run.



To understand how the economy works in the short run, we need a new model. Our new model focuses on how real and nominal variables interact.

The model of Aggregate Demand and Aggregate supply.



The model of Aggregate Demand and Aggregate supply

The model of short run economic fluctuations focus on the behavior of two variables. The first variable is the economy's output of goods and services, as measured by real GDP. The second is the average level of prices, as measured by the CPI or GDP deflator.

Notice that output is a real variable, where as the price level is a nominal variable. By focusing on the relationship between these



Two variables, we are departing from the classical assumption that real and nominal variables can be studied separately.

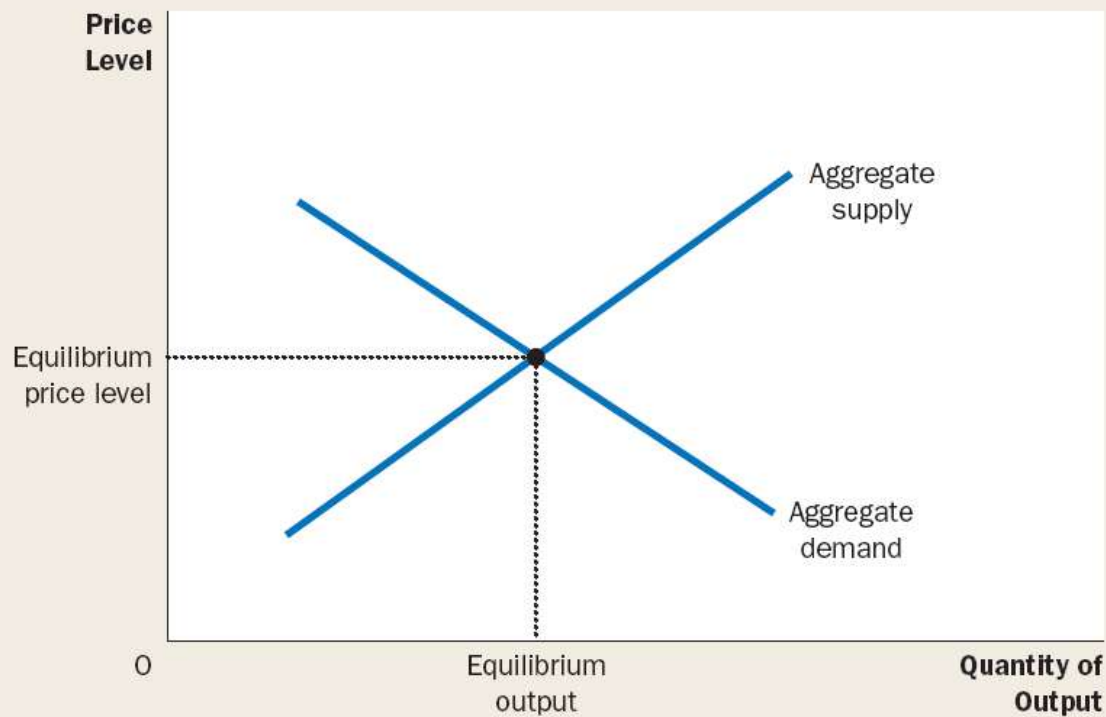


FIGURE 2

Aggregate Demand and Aggregate Supply

Economists use the model of aggregate demand and aggregate supply to analyze economic fluctuations. On the vertical axis is the overall level of prices. On the horizontal axis is the economy's total output of goods and services. Output and the price level adjust to the point at which the aggregate-supply and aggregate-demand curves intersect.



The aggregate demand curve shows the quantity of goods and services that households, firms, the govt, and customers abroad want to buy at each price level.

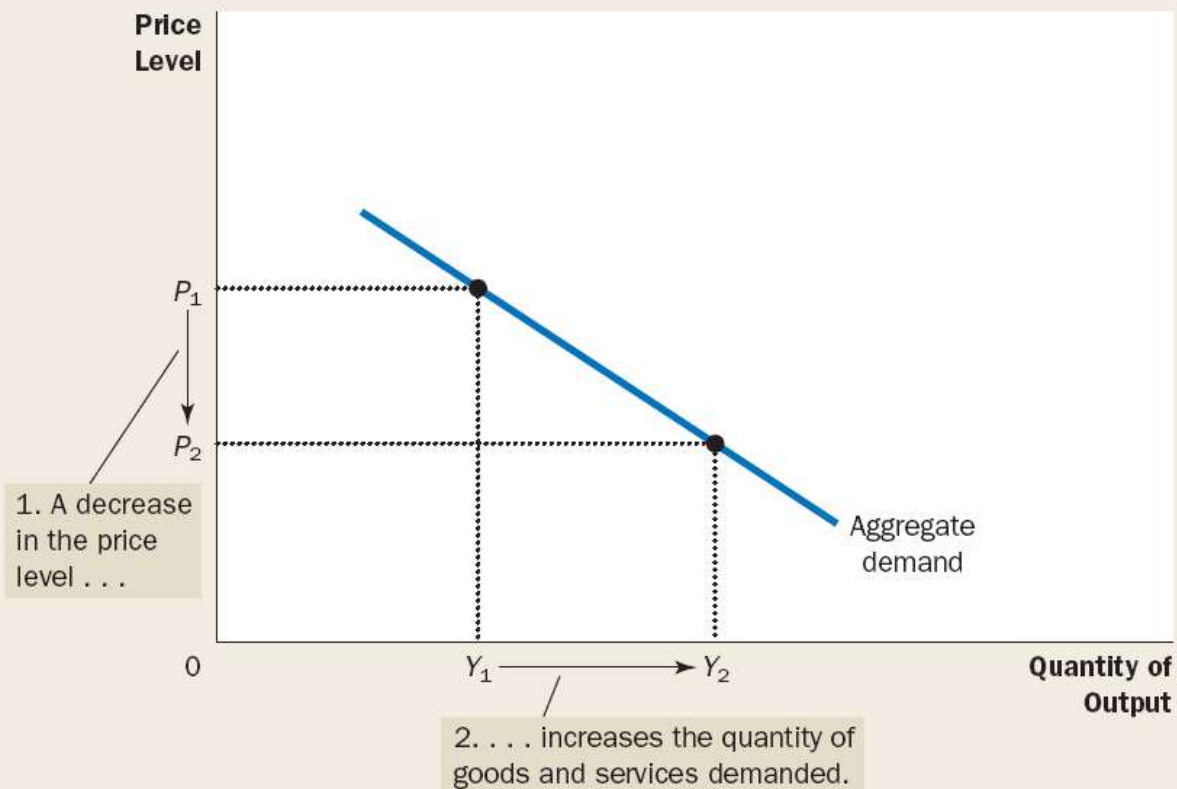
The aggregate supply curve shows natural rate of output and price level.

3

FIGURE

The Aggregate-Demand Curve

A fall in the price level from P_1 to P_2 increases the quantity of goods and services demanded from Y_1 to Y_2 . There are three reasons for this negative relationship. As the price level falls, real wealth rises, interest rates fall, and the exchange rate depreciates. These effects stimulate spending on consumption, investment, and net exports. Increased spending on any or all of these components of output means a larger quantity of goods and services demanded.





Price Level and Consumption

$P \downarrow \rightarrow$ Real value of money $\uparrow$ and consumers wealthier, which encourage them to spend more in goods and services.

$\uparrow$ Consumer spending means larger QD.

Conversely,

$P \uparrow \rightarrow$ Real value of money $\downarrow \rightarrow$ Wealth $\downarrow \rightarrow$
Consumer spending means smaller QD $\downarrow$



The Price level and Investment

The price level is one determinant of the quantity of money demand.

$P \downarrow$ (lower price) $\rightarrow$ less money household needs to buy goods and services they want $\rightarrow$ Household try to reduce their holdings of money by **lending some of it out means to buy interest-bearing bonds or deposit it excess money in an interest bearing saving accounts, and bank would use these funds to make more loans.** In either case, as households try to convert some of their money into interest-bearing assets, they drive down interest rate ($\downarrow$). $\rightarrow$ borrowing less expensive $\rightarrow$ borrow more to invest in new plants, equipments (firms) $\uparrow \rightarrow$ house hold borrow more to invest in new housing, consumer durables (which are bought on credit) $\uparrow$.



This gives the second reason the aggregate demand curve is downward sloping.

$P \downarrow \rightarrow r.o.i \downarrow \rightarrow$ encourage greater spending on investment goods, and thereby increases the quantity of goods and services demanded $\uparrow$. Conversely,

$P \uparrow \rightarrow r.o.i \uparrow \rightarrow$ reduce the investment spending and the quantity of goods and services demanded $\downarrow$.



The Price Level and Net Exports

$P \downarrow \rightarrow r.o.i \downarrow$ → in response to lower r.o.i, some Indian investors will seek higher returns by investing abroad. For instance, as the r.o.i on Indian govt bonds falls, **a mutual fund might sell Indian govt bonds to buy German bonds**. As the mutual fund tries to convert its rupees into euros to buy the German bonds, it **increases the supply of rupees** in the market for foreign-currency exchange. The $\uparrow$ ss of Rupees to be turned into euros causes the **rupees to depreciate** relative to the euro. This leads to a change in the real exchange rate- the relative price of domestic and foreign goods. Because each rupees buys fewer units of foreign currencies, **foreign goods becomes more expensive relative to domestic goods**.

The change in relative prices affects spending, both at home and abroad. Because foreign goods are more expensive, Indian buys less from other countries, causing **India imports of goods and services to decrease**. At the same time, because **India goods are now cheaper, foreigners buy more from India, so India exports increase**. Net exports equal exports minus imports, so both of these changes cause **India net exports to $\uparrow$ increase**. Thus, the fall in the real exchange value of rupees leads to an increase in QD.



TABLE 1

The Aggregate-Demand Curve: Summary

Why Does the Aggregate-Demand Curve Slope Downward?

1. *The Wealth Effect:* A lower price level increases real wealth, which encourages spending on consumption.
2. *The Interest-Rate Effect:* A lower price level reduces the interest rate, which encourages spending on investment.
3. *The Exchange-Rate Effect:* A lower price level causes the real exchange rate to depreciate, which encourages spending on net exports.

Why Might the Aggregate-Demand Curve Shift?

1. *Shifts Arising from Consumption:* An event that makes consumers spend more at a given price level (a tax cut, a stock-market boom) shifts the aggregate-demand curve to the right. An event that makes consumers spend less at a given price level (a tax hike, a stock-market decline) shifts the aggregate-demand curve to the left.
2. *Shifts Arising from Investment:* An event that makes firms invest more at a given price level (optimism about the future, a fall in interest rates due to an increase in the money supply) shifts the aggregate-demand curve to the right. An event that makes firms invest less at a given price level (pessimism about the future, a rise in interest rates due to a decrease in the money supply) shifts the aggregate-demand curve to the left.
3. *Shifts Arising from Government Purchases:* An increase in government purchases of goods and services (greater spending on defense or highway construction) shifts the aggregate-demand curve to the right. A decrease in government purchases on goods and services (a cutback in defense or highway spending) shifts the aggregate-demand curve to the left.
4. *Shifts Arising from Net Exports:* An event that raises spending on net exports at a given price level (a boom overseas, speculation that causes an exchange-rate depreciation) shifts the aggregate-demand curve to the right. An event that reduces spending on net exports at a given price level (a recession overseas, speculation that causes an exchange-rate appreciation) shifts the aggregate-demand curve to the left.



It is important to keep in mind that the aggregate-demand curve (like all demand curves) is drawn holding “other things equal.” In particular, our three explanations of downward-sloping aggregate-demand curve assume that money supply is fixed.

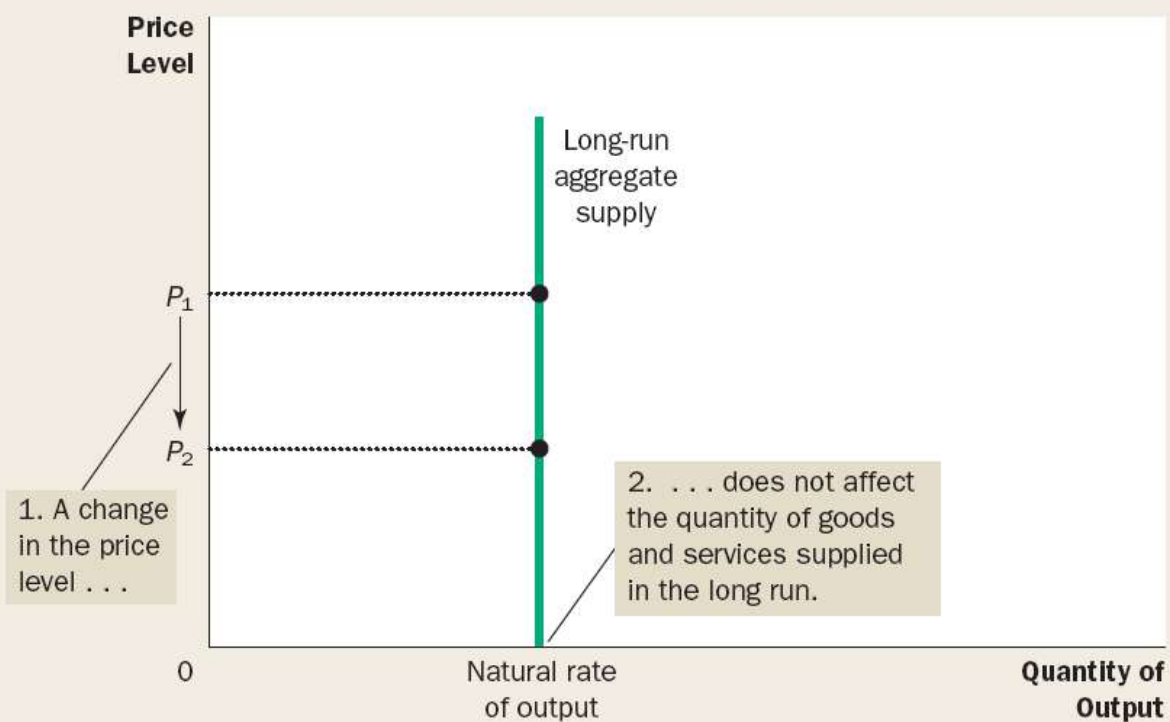
We will see in the next chapter, a change in the quantity of money shifts the aggregate-demand curve. At this point just keep in mind that the aggregate-demand curve is drawn for a given quantity of money.

4

FIGURE

The Long-Run Aggregate-Supply Curve

In the long run, the quantity of output supplied depends on the economy's quantities of labor, capital, and natural resources and on the technology for turning these inputs into output. Because the quantity supplied does not depend on the overall price level, the long-run aggregate-supply curve is vertical at the natural rate of output.





Why the Aggregate-Supply Curve is Vertical in the Long Run ?

In the long run, an economy's production of goods and services (its real GDP) depends on its supplies of labour, capital, and natural resources and on the available technology used to turn these factors of production into goods and services.

The vertical L-R aggregate supply curve is a graphical representation of classical dichotomy and monetary neutrality. Most economists believe that this principle works well when studying the economy over a period of many years but not when studying year – to – year changes.

A background image showing two construction workers in blue shirts and hard hats working on a wooden roof structure. In the background, there is a red brick building with white-framed windows.

Why the Aggregate-Supply Curve is Vertical in the Long Run ?

- The long run level of production is sometimes called *potential output* or *full-employment output*. To be more precise, it is the natural rate of output because it shows what the economy produces when unemployment is at its natural, or normal, rate. The natural rate of output is the level of production towards which the economy gravitates in the long run.



Shifts in the long run aggregate supply curve

Shift Arising from Changes in Labour:

Imagine that an economy experiences an increase in immigration. There would be a greater number of workers, the quantity of goods and services supply would increase. As a result, the long run aggregate supply curve would shift to the right. Conversely, if many workers left the economy to go abroad, the L-R aggregate supply curve would shift to the left.



- Shift Arising from Changes in Capital

An increase in the economy's capital stock increases productivity, thereby, the quantity of goods and services supplied. As a result, the long-run aggregate supply curve shifts to the right.

Conversely, a decrease in the economy's capital stock decreases productivity and the quantity of goods and services supplied, shifting the L-R aggregate supply curve to the left.



Long run aggregate supply curve shifts arising from changes in **natural resources**.

As the production depends on the natural resources, including land, minerals and weather. A discovery of new mineral deposit shifts the long run aggregate-supply curve to the right.

A change in weather patterns that makes farming more difficult shifts the L-R aggregate-supply curve to the left.



Long run aggregate supply curve shifts arising from changes in **technological knowledge**. The invention of computer, for instance, has allowed us to produce more goods and services from any given amounts of labour, capital, and natural resources. As the computer use has spread throughout the economy, it has shifted the L-R aggregate supply curve to right.



Long-run aggregate curve reflects the classical model of the economy. Any policy or event that raised real GDP can be described as increasing the quantity of goods and services supplied and shifting the long run aggregate supply curve to the right.

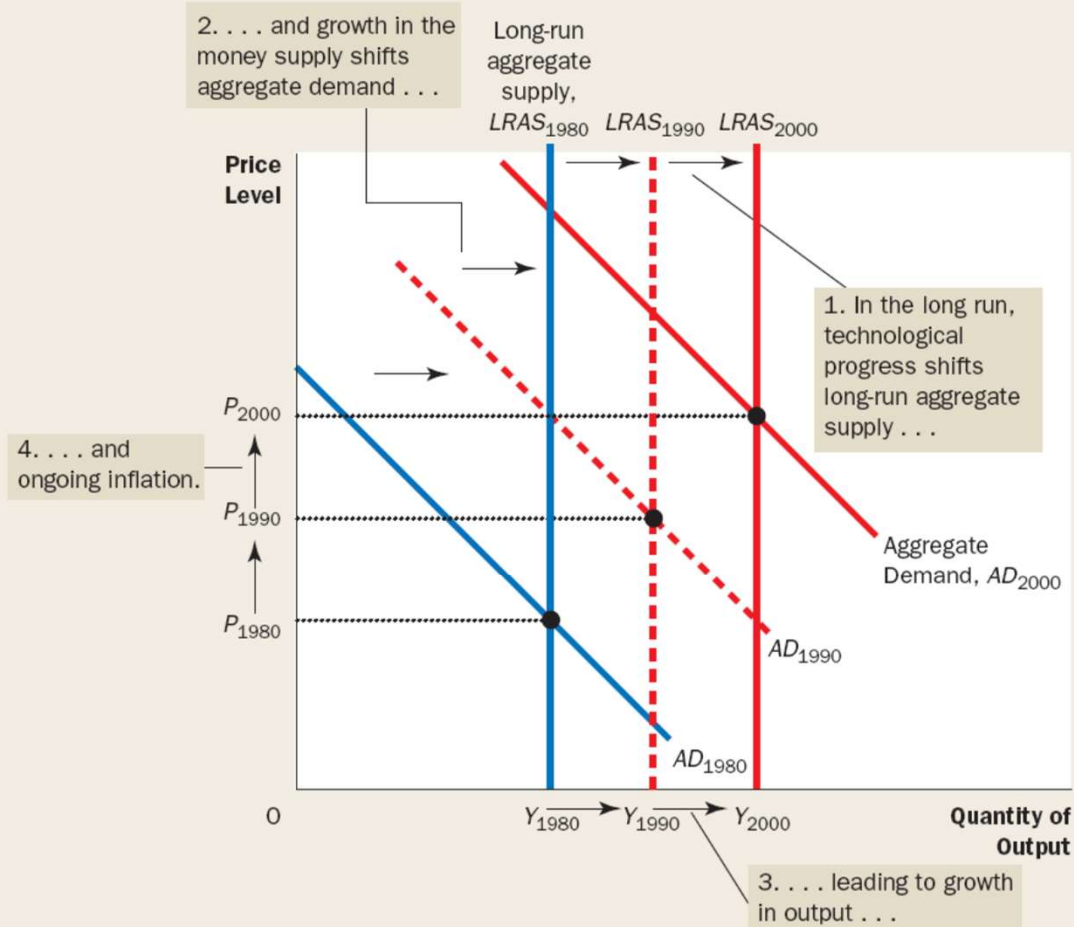
Any policy or event that lowered real GDP can be described as decreasing the quantity of goods and services supplied and shifting the long run AS curve to the left.

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FIGURE

Long-Run Growth and Inflation in the Model of Aggregate Demand and Aggregate Supply

As the economy becomes better able to produce goods and services over time, primarily because of technological progress, the long-run aggregate-supply curve shifts to the right. At the same time, as the Fed increases the money supply, the aggregate-demand curve also shifts to the right. In this figure, output grows from Y_{1980} to Y_{1990} and then to Y_{2000} , and the price level rises from P_{1980} to P_{1990} and then to P_{2000} . Thus, the model of aggregate demand and aggregate supply offers a new way to describe the classical analysis of growth and inflation.

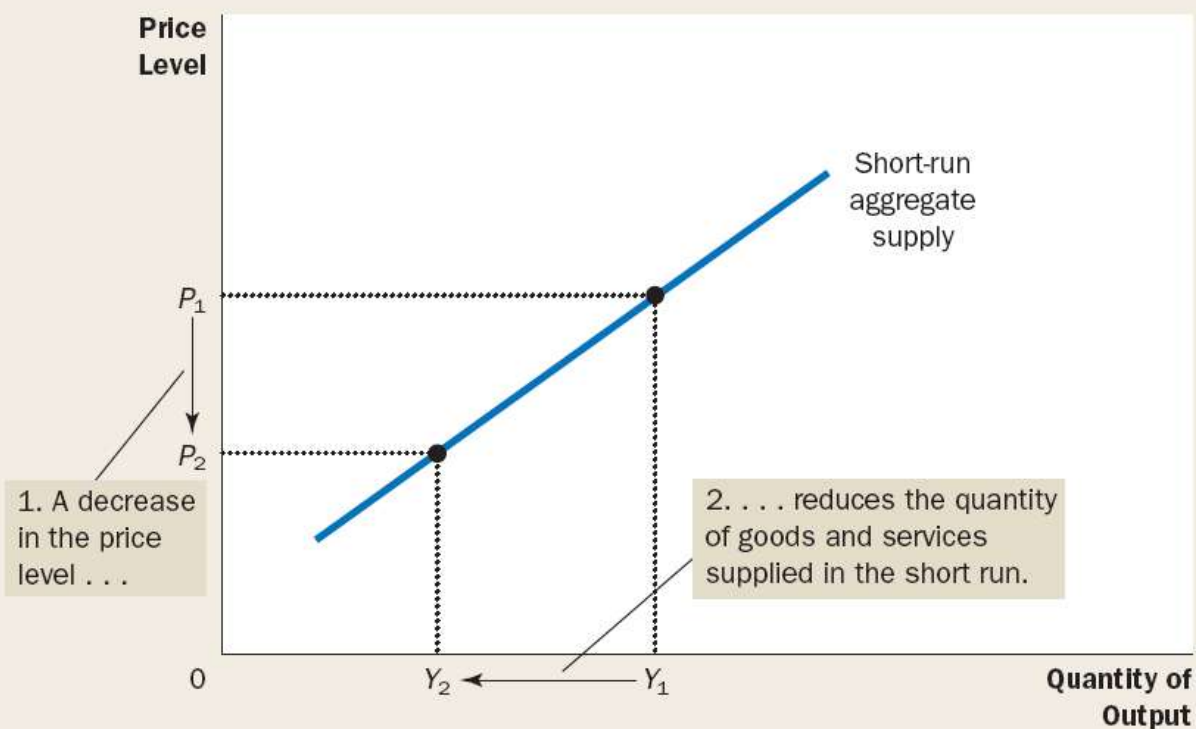


6

FIGURE

The Short-Run Aggregate-Supply Curve

In the short run, a fall in the price level from P_1 to P_2 reduces the quantity of output supplied from Y_1 to Y_2 . This positive relationship could be due to sticky wages, sticky prices, or misperceptions. Over time, wages, prices, and perceptions adjust, so this positive relationship is only temporary.





Sticky Wage Theory

According to this theory SR Aggregate Supply curve slopes upward because nominal wages are slow to adjust to changing economic condition.

When firm prices are fall below expectations, the firm get less price than expected for each unit of product it sells. As the production is now less profitable, so the firm higher fewer workers and reduces the quantity of output supplied. Over time, labour contract will expire and firm will renegotiate with it workers for a lower wage (which they may accept because of lower prices).

The top of the slide features a horizontal banner image. It is a painting of a red building with several windows. Two figures are visible on a roof or ledge in the background. The style is somewhat impressionistic.

The Sticky-Price Theory

- This theory emphasizes that the price of some goods and services also adjust sluggishly in response to changing economic conditions. The slow adjustment of prices occur because there are cost to adjusting prices, called menu costs. These menu costs include the cost of printing and distributing catalogs and the time required to change price tags. As a result of these costs, prices as well as wages may be sticky in the short run.

2

T A B L E

The Short-Run Aggregate-Supply Curve: Summary

Why Does the Short-Run Aggregate-Supply Curve Slope Upward?

1. *The Sticky-Wage Theory*: An unexpectedly low price level raises the real wage, which causes firms to hire fewer workers and produce a smaller quantity of goods and services.
2. *The Sticky-Price Theory*: An unexpectedly low price level leaves some firms with higher-than-desired prices, which depresses their sales and leads them to cut back production.
3. *The Misperceptions Theory*: An unexpectedly low price level leads some suppliers to think their relative prices have fallen, which induces a fall in production.

Why Might the Short-Run Aggregate-Supply Curve Shift?

1. *Shifts Arising from Labor*: An increase in the quantity of labor available (perhaps due to a fall in the natural rate of unemployment) shifts the aggregate-supply curve to the right. A decrease in the quantity of labor available (perhaps due to a rise in the natural rate of unemployment) shifts the aggregate-supply curve to the left.
2. *Shifts Arising from Capital*: An increase in physical or human capital shifts the aggregate-supply curve to the right. A decrease in physical or human capital shifts the aggregate-supply curve to the left.
3. *Shifts Arising from Natural Resources*: An increase in the availability of natural resources shifts the aggregate-supply curve to the right. A decrease in the availability of natural resources shifts the aggregate-supply curve to the left.
4. *Shifts Arising from Technology*: An advance in technological knowledge shifts the aggregate-supply curve to the right. A decrease in the available technology (perhaps due to government regulation) shifts the aggregate-supply curve to the left.
5. *Shifts Arising from the Expected Price Level*: A decrease in the expected price level shifts the short-run aggregate-supply curve to the right. An increase in the expected price level shifts the short-run aggregate-supply curve to the left.

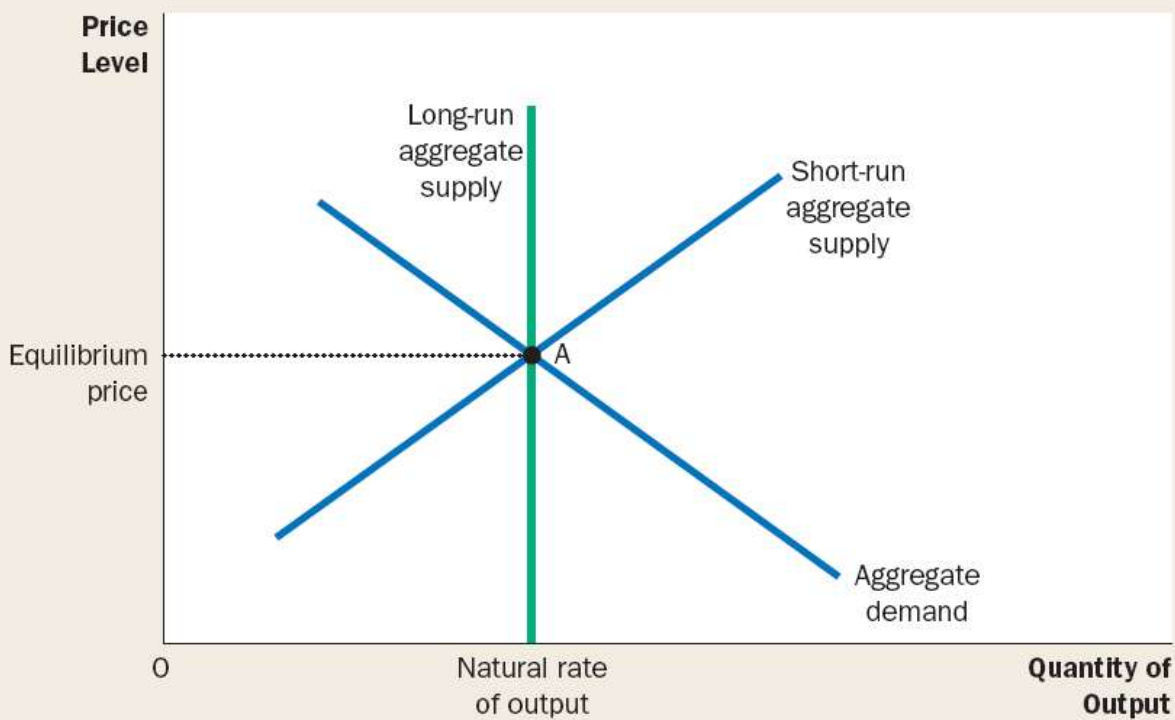


FIGURE 7

The Long-Run Equilibrium

The long-run equilibrium of the economy is found where the aggregate-demand curve crosses the long-run aggregate-supply curve (point A). When the economy reaches this long-run equilibrium, the expected price level will have adjusted to equal the actual price level. As a result, the short-run aggregate-supply curve crosses this point as well.



TABLE 3

Four Steps for Analyzing Macroeconomic Fluctuations

1. Decide whether the event shifts the aggregate demand curve or the aggregate supply curve (or perhaps both).
2. Decide in which direction the curve shifts.
3. Use the diagram of aggregate demand and aggregate supply to see how the shift changes output and the price level in the short run.
4. Use the diagram of aggregate demand and aggregate supply to analyze how the economy moves from its new short-run equilibrium to its long-run equilibrium.

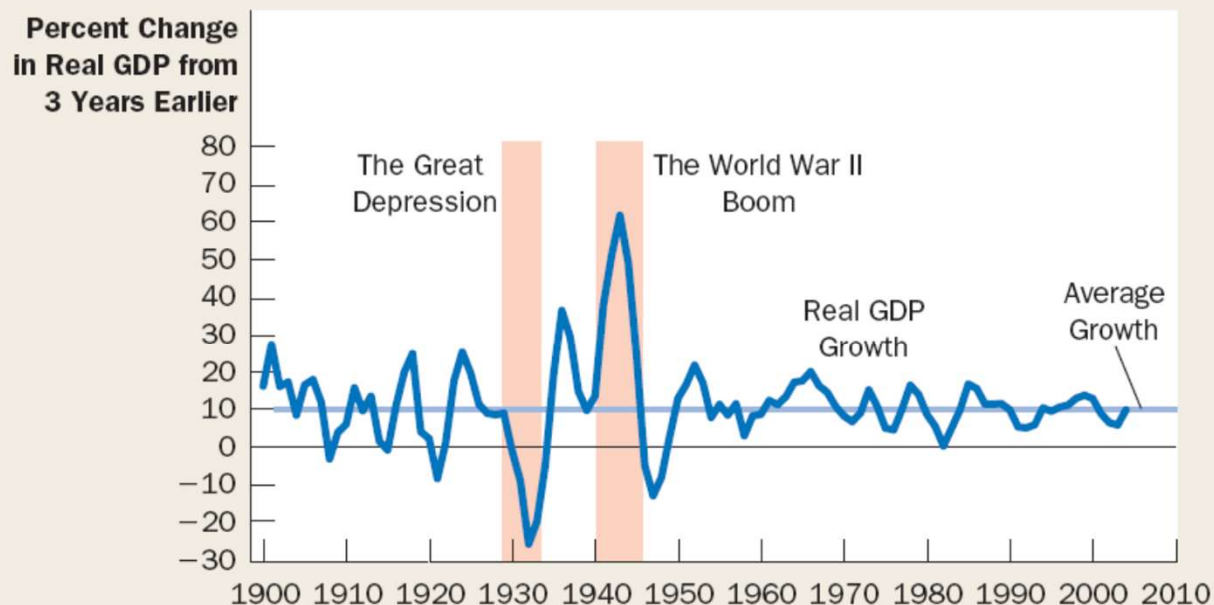
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FIGURE

**U.S. Real GDP
Growth since 1900**

Source: U.S. Department of
Commerce.

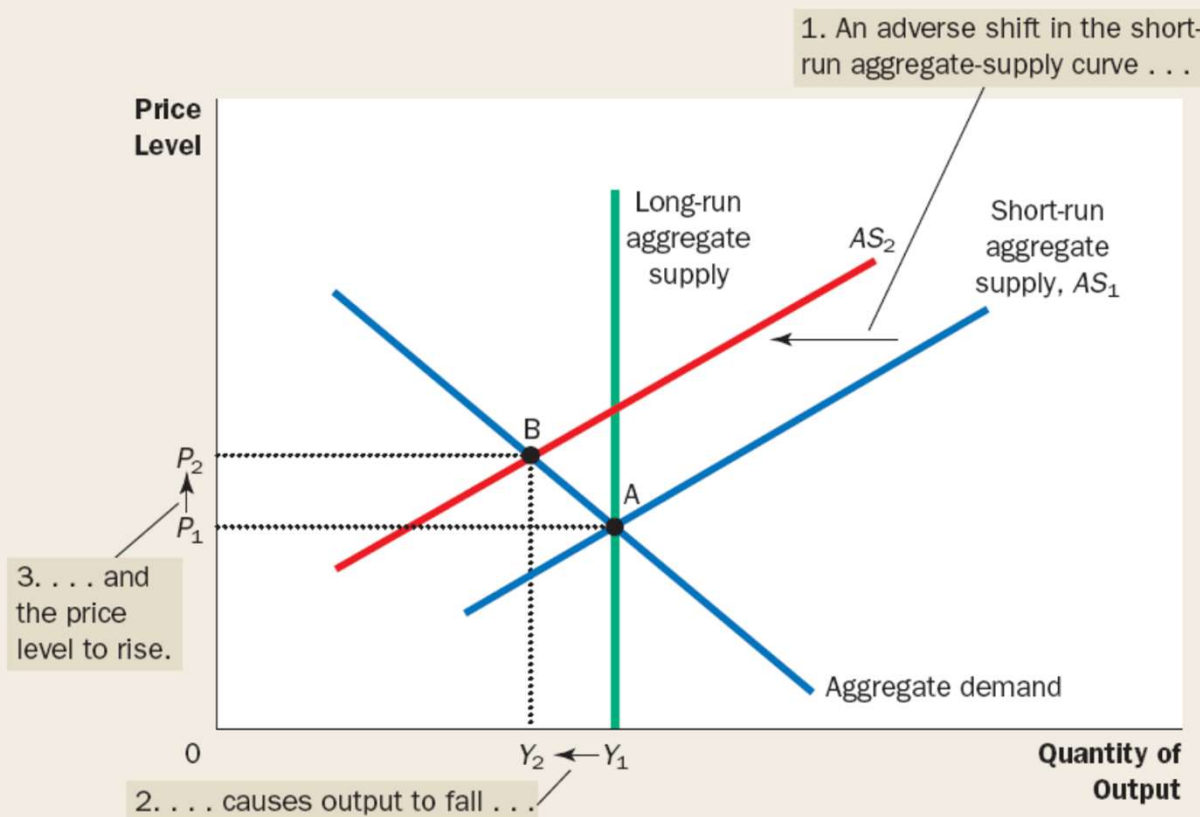
Over the course of U.S. economic history, two fluctuations stand out as especially large. During the early 1930s, the economy went through the Great Depression, when the production of goods and services plummeted. During the early 1940s, the United States entered World War II, and the economy experienced rapidly rising production. Both of these events are usually explained by large shifts in aggregate demand.



10 FIGURE

An Adverse Shift in Aggregate Supply

When some event increases firms' costs, the short-run aggregate-supply curve shifts to the left from AS_1 to AS_2 . The economy moves from point A to point B. The result is stagflation: Output falls from Y_1 to Y_2 , and the price level rises from P_1 to P_2 .



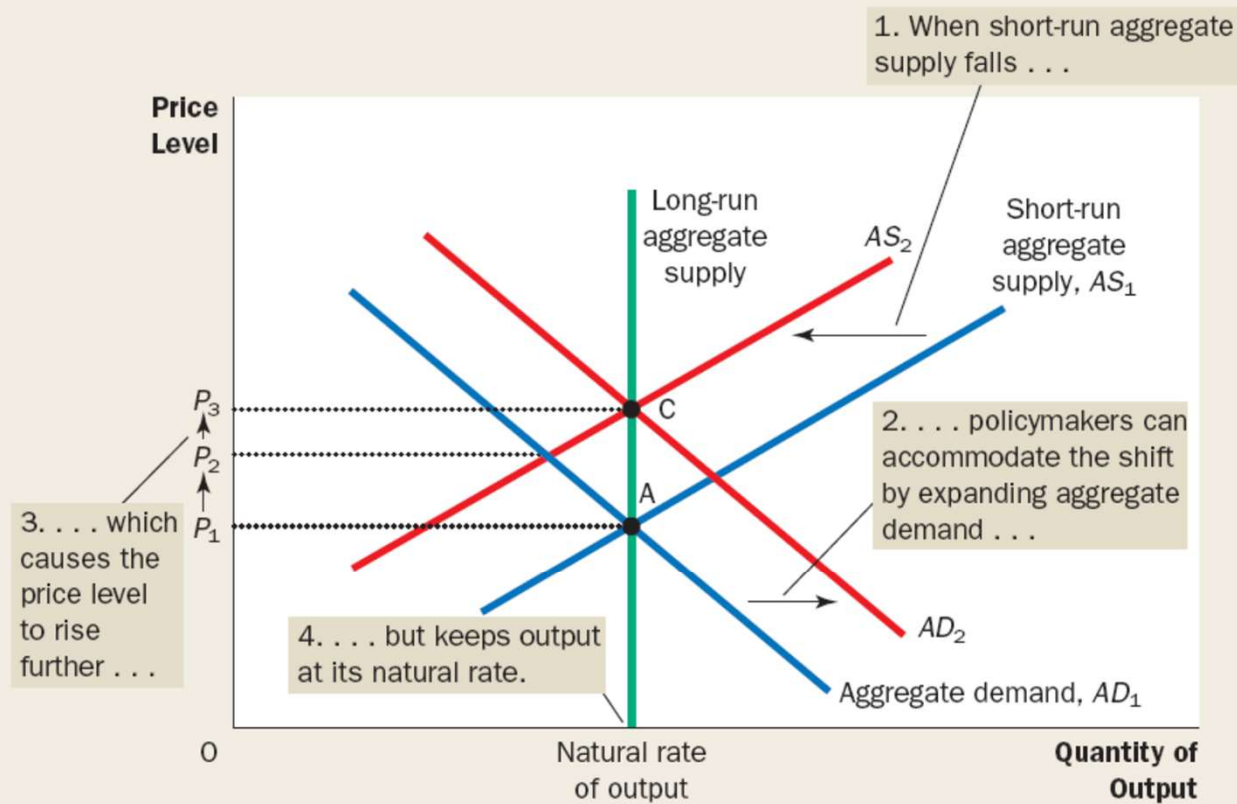


FIGURE 11

Accommodating an Adverse Shift in Aggregate Supply

Faced with an adverse shift in aggregate supply from AS_1 to AS_2 , policymakers who can influence aggregate demand might try to shift the aggregate-demand curve to the right from AD_1 to AD_2 . The economy would move from point A to point C. This policy would prevent the supply shift from reducing output in the short run, but the price level would permanently rise from P_1 to P_3 .